AN ETALE NEIGHBOURHOOD OF A POINT OF A FIBRE LIFTS TO THE TOTAL SPACE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10246-10256 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 10258-10276, the complete author proof (17 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Etale neighbourhoods at a point, fibres of a morphism, and open immersions; the construction is retained.

Source reference. Original lemma 00U2: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10276–10276. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04D1: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10276–10276. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Lift étale neighbourhood of point on fibre to total space. Let $X \to S$ be a morphism of schemes. Let $x \in X$ with image $s \in S$. Let $(V, v) \to (X_s, x)$ be an étale neighbourhood. Then there exists an étale neighbourhood $(U, u) \to (X, x)$ such that there exists a morphism $(U_s, u) \to (V, v)$ of étale neighbourhoods of (X_s, x) which is an open immersion.*

Proof. We may assume X, V, and S affine. Say the morphism $X \to S$ is given by $A \to B$ the point x by a prime $\mathfrak{q} \subset B$, the point s by $\mathfrak{p} = A \cap \mathfrak{q}$, and the morphism $V \to X_s$ by $B \otimes_A \kappa(\mathfrak{p}) \to C$. Since $\kappa(\mathfrak{p})$ is a localization of $A/\mathfrak{p}$ there exists an $f \in A$, $f \notin \mathfrak{p}$ and an étale ring map $B \otimes_A (A/\mathfrak{p})_f \to D$ such that

$$C = (B \otimes_A \kappa(\mathfrak{p})) \otimes_{B \otimes_A (A/\mathfrak{p})_f} D$$

See Algebra, Lemma 00U2 part (9). After replacing A by A_f and B by B_f we may assume D is étale over $B \otimes_A A/\mathfrak{p} = B/\mathfrak{p}B$. Then we can apply Algebra, Lemma 04D1. This proves the lemma. □

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